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Day 4: DHT11 Temperature & Humidity Sensor

- **Date:** 22-07-2025
- **Domain:** Embedded Systems and IoT
- **Topic:** Study and Implementation of DHT11 Sensor with Arduino Uno

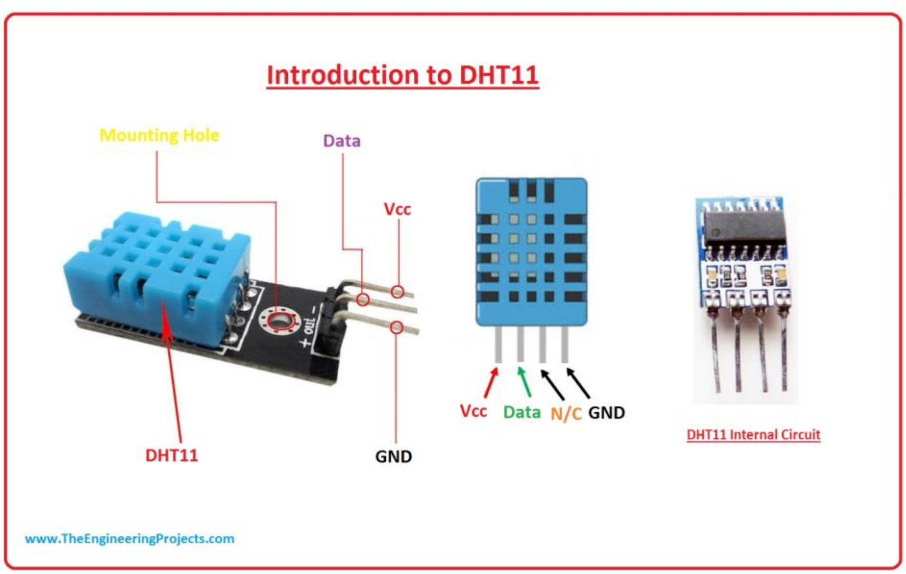
Objective

To understand the working of the DHT11 temperature and humidity sensor, explore its technical specifications, analyze its datasheet, and implement sensor data reading using Arduino Uno and the appropriate library.

What is DHT11?

The DHT11 is a basic, low-cost digital sensor that measures temperature and humidity. It uses a capacitive humidity sensor and a thermistor to measure the surrounding air and provides a digital output.

It is commonly used in environmental monitoring, IoT weather stations, smart agriculture, and automation systems.



DHT11 Key Features

Feature	Specification
Temperature Range	0°C to 50°C
Humidity Range	20% to 90% RH
Temperature Accuracy	±2°C
Humidity Accuracy	±5% RH
Operating Voltage	3.3V – 5.5V
Signal Output	Digital (Single-wire Serial Communication)
Sampling Rate	1 Hz (1 reading per second)

Pin Configuration

Pin	Name	Description
1	VCC	Power supply (3.3V – 5V)
2	DATA	Serial data output
3	NC	Not connected (sometimes omitted)
4	GND	Ground

Note: In some 3-pin modules, pin 3 is skipped.

DHT11 Datasheet Summary

- Contains a calibrated digital signal output.
- Requires a pull-up resistor (typically 10KΩ) on the data pin.
- Communication uses a single-wire protocol with precise timing.
- Response time: <5 seconds
- Lifetime: >50000 measurements
- Low power consumption: 0.3mW (when measuring)

Advantages

- Low cost and easy to use
- Requires only one GPIO pin
- Works with both 3.3V and 5V systems
- Supported by various libraries in Arduino and Python

Disadvantages

- Limited accuracy compared to DHT22
- Slow sampling rate (1Hz)
- Not suitable for high-precision applications
- Fixed signal protocol — timing critical

Limitations

- Cannot measure temperatures below 0°C or above 50°C
- Humidity range limited to 20–90% RH
- Slower refresh rate (1 reading per second only)
- May degrade in extreme environmental conditions over time

Required Library

To interface DHT11 with Arduino, the “DHT sensor library” by Adafruit is commonly used.

How to Install:

1. Open Arduino IDE
2. Go to Sketch > Include Library > Manage Libraries

3. Search for "DHT sensor library" by Adafruit and install it
4. It may also install the Adafruit Unified Sensor Library as a dependency

Arduino Wiring

DHT11 Pin	Arduino Pin
VCC	5V
DATA	Digital 2
GND	GND

Use a 10KΩ pull-up resistor between DATA and VCC.

Sample Arduino Code

```
#include "DHT.h"


#define DHTPIN 2          // Digital pin connected to the DHT sensor
#define DHTTYPE DHT11     // DHT11 Sensor


DHT dht(DHTPIN, DHTTYPE);


void setup() {
  Serial.begin(9600);
  dht.begin();
}


void loop() {
  float temp = dht.readTemperature();    // Read temperature in Celsius
```

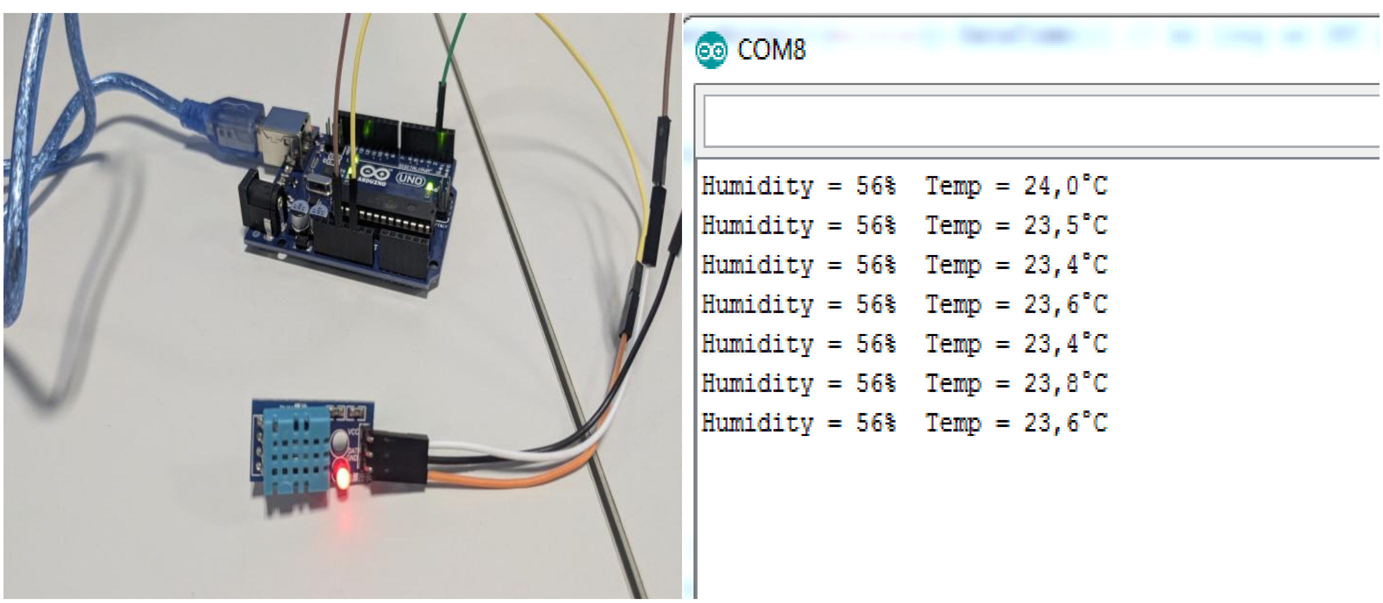
```
  float hum = dht.readHumidity();        // Read humidity


  if (isnan(temp) || isnan(hum)) {
    Serial.println("Failed to read from DHT11 sensor!");
    return;
  }


  Serial.print("Temperature: ");
  Serial.print(temp);
  Serial.print(" °C, Humidity: ");
  Serial.print(hum);
  Serial.println(" %");


  delay(2000);  // Wait 2 seconds before next reading
}
```

Output:



Result

- DHT11 successfully connected with Arduino Uno.
- Temperature and humidity values printed on the Serial Monitor every 2 seconds.
- Verified working within expected operating range.

Learning Outcomes

- Understood how digital sensors like DHT11 work.
- Learned about single-wire serial communication protocol.
- Installed and used external Arduino libraries.
- Practiced interfacing sensors with Arduino Uno.
- Observed real-time environmental data from hardware.